

# Borg (Bring yer Own GRaphics): An Open-Source Tile-Based GPU with Silicon Tapeout

Andreas Wendleder

Independent Researcher, Landshut, Germany   [andreas.wendleder@proton.me](mailto:andreas.wendleder@proton.me)   [github.com/gonsolo/Borg](https://github.com/gonsolo/Borg)  
[orcid.org/0009-0008-8304-2818](https://orcid.org/0009-0008-8304-2818)

HPG 2026, July 17–19, Los Angeles, CA

## Motivation

- Every commercial GPU hides its microarchitecture behind NDAs
- This blocks **reproducibility**, **teaching**, and architectural experiments
- Concise:** Hardware (**1513 LoC** Scala), Compiler (**1600 LoC** Python), Firmware (**2000 LoC** C). Total stack **< 6k lines**.
- Open GPU simulators exist — but *no open GPU has reached silicon*

**Borg** is the fully open answer: a TBDR GPU SoC in Chisel, built with an open toolchain, and **taped out twice on real silicon**. Borg has contributed **68 patches upstream** to the open EDA ecosystem. Integrating the **TinyQV** RISC-V core (by Michael Bell), Borg provides a complete system-level open-source hardware reference.

To our knowledge: the only open-source GPU with a completed ASIC tapeout.

## Development Roadmap: Phases 1–6

- Phase 1** ✓ — **Foundational Architecture** (2025)
- Phase 2** [wip] — **GPU Autonomy / 28-Day Sprint**
- Phase 3** — **Linux-Capable CPU** (M/A/MMU)
- Phase 4** — **Mesa Vulkan Driver** (NIR → SPIR-B)
- Phase 5** — **Hardware Extensions** (SIMD, Blending)
- Phase 6** — **Vulkan 1.0 Conformance**

### Status Key Phase 2 Milestones

- ✓ 32-bit RISC-V ISA + Perspective Proj.
- ✓ Triangle Clipping + Fragment Interpolation
- ✓ Pixel Iterator (Rasterizer) + Z-Buffer
- ✓ Command FIFO + SystemRDL Register Map
- ✓ Texture Fetch (PSRAM) + Shared Mem. Ctrl
- [wip] GPU DMA Engine + VGA Output

## Design Values

**Transparency** No binary blobs, no hidden firmware.  
**Accessibility** Runs on a \$20 FPGA; uses \$500 ASIC runs.  
**Simplicity** Minimalist TBDR pipeline for maximum performance/area.

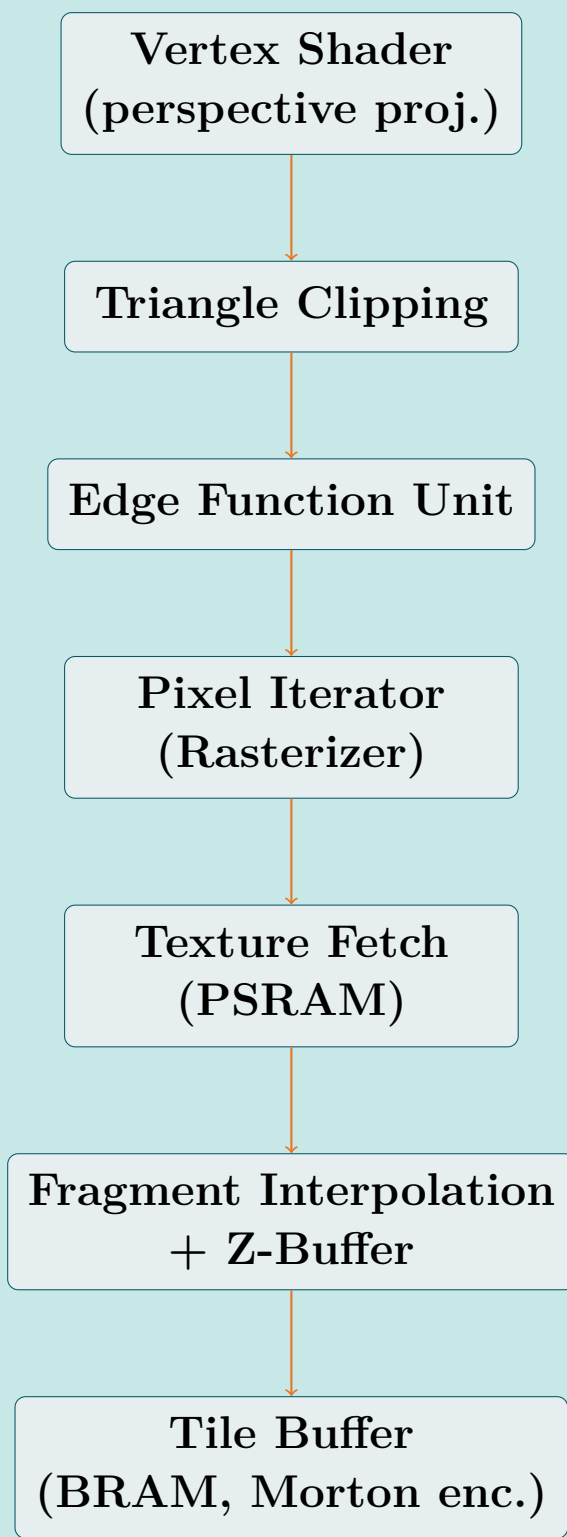
## The Transparency Gap: Borg vs. Commercial

| Feature        | Borg          | Mobile      | Desktop     |
|----------------|---------------|-------------|-------------|
| RTL Source     | <b>Open</b>   | Proprietary | Proprietary |
| EDA Toolchain  | <b>Libre</b>  | Proprietary | Proprietary |
| Silicon Layout | <b>Public</b> | Secret      | Secret      |
| ISA Specs      | <b>Public</b> | Often NDA   | Proprietary |

**Borg** is the only design where researchers can inspect the layout and modify logic without NDAs.

## SoC & TBDR Architecture

Borg is a complete **RISC-V SoC** featuring the **TinyQV** core. The GPU follows the Tile-Based Deferred Rendering paradigm:



Geometry is processed tile-by-tile; shaded fragments are written to an on-chip tile buffer, minimising external memory bandwidth.

## ASIC Tapeouts (Tiny Tapeout)

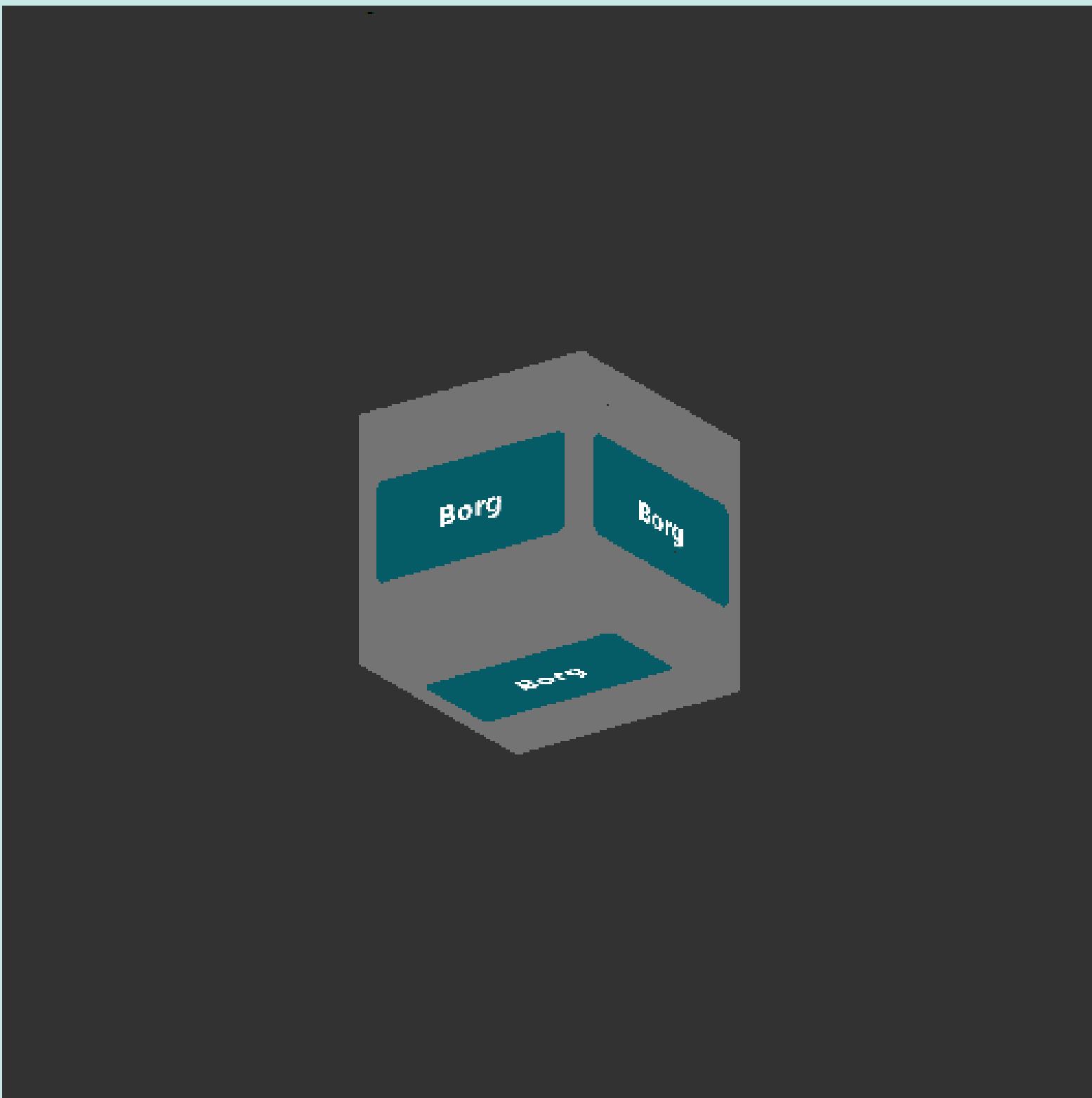
| Run      | Process    | Tiles    | Content       | Cost   |
|----------|------------|----------|---------------|--------|
| TTIHP26a | IHP 130 nm | 8 tiles  | FP peripheral | 515 €  |
| TTSKY26a | SKY 130 nm | 16 tiles | Rasteriser    | 1235 € |

Chips for TTIHP26a expected July 2026.

## Current Demo

Running on **pico-ice** (iCE40 UP5K, 5280 LCs):

- 12 textured triangles, 32×32 framebuffer
- Full pipeline: vertex shader → texture fetch → Z-buffer → output
- Orchestrated by integrated **TinyQV** RISC-V core



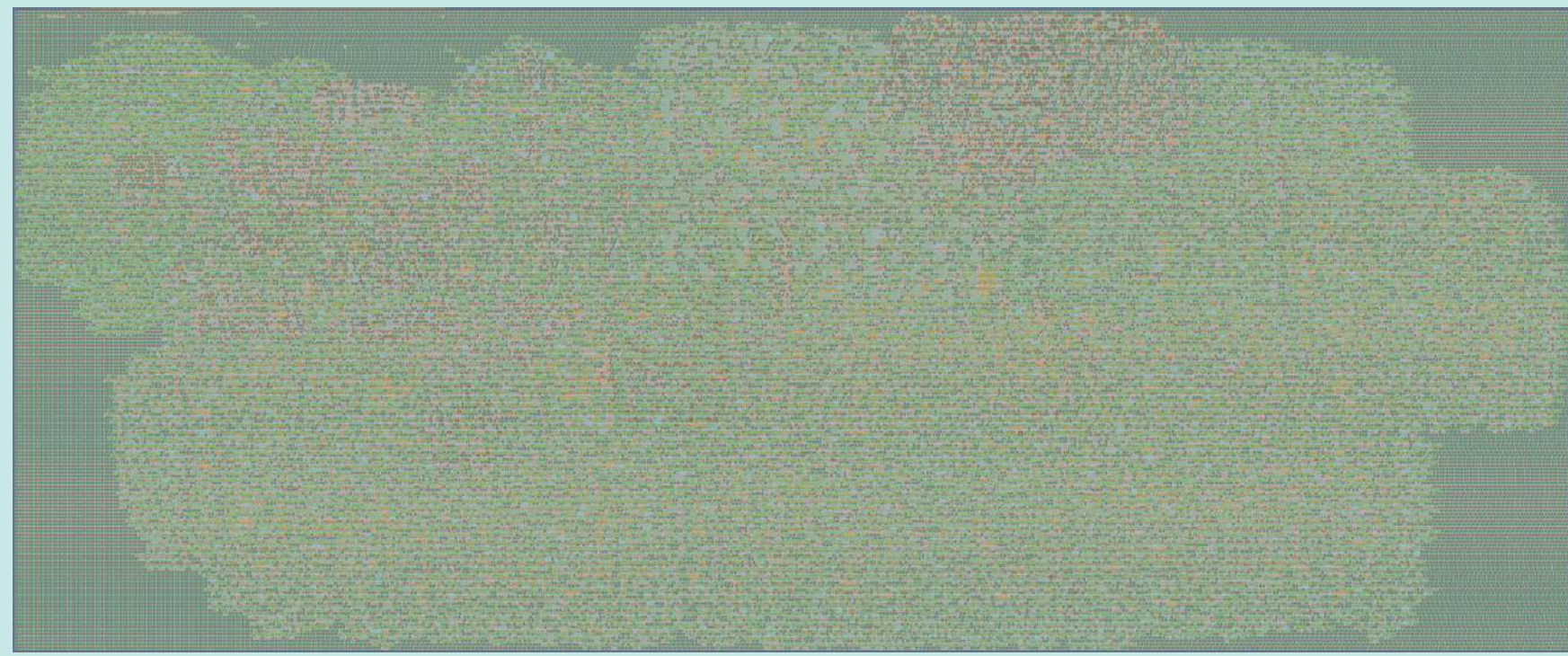
VkCube running in Verilator simulation (500×500)

**Bandwidth note:** Current bottleneck is shared PSRAM access. GPU DMA engine (Phase 3) will eliminate CPU involvement.

## Chisel: Hardware as Code

- High Productivity:** 28-day sprint made possible by Scala’s expressiveness compared to traditional Verilog/VHDL.
  - Type Safety:** FIRRTL compiler catches wiring errors early.
- “Borg proves that a single researcher can build a complete GPU pipeline in weeks.”

## Physical Design (GDSII)



| Metric          | Value (TTSKY26a)        |
|-----------------|-------------------------|
| Logic Area      | 1.4 × 1.4 mm (16 tiles) |
| Est. Gate Count | ~180k (excluding BRAM)  |
| Clock Target    | 50 MHz (Standard Cell)  |

## FPGA Resource Budget (iCE40 UP5K)

| Resource     | Used   | Available |
|--------------|--------|-----------|
| Logic Cells  | ~5 280 | 5 280     |
| BRAMs (4 kb) | ~14    | 30        |
| DSPs         | 1      | 8         |

## Open-Source Toolchain

**HDL** Chisel (Scala, **1513 lines**)  
**Synthesis** Yosys  
**P&R** nextpnr-ice40 / **LibreLane** (Nix)  
**Simulation** Verilator, cocotb, Arcilator  
**Compiler** SPIR-V → SPIR-B (**1600 lines** Python)  
**Firmware** 32-bit RISC-V (**2000 lines** C)  
**Reg. map** PeakRDL-chisel  
**Math Cores** **Berkeley Hardfloat** (FMA)

**Reproducible Build:** `direnv` + `Nix` flakes.  
Zero-install setup for all 18+ EDA tools.

**Ecosystem Impact:** Pioneers the first fully reproducible ASIC flow in `nixpkgs` (LibreLane merged Jan 2026).

## Future Work & Open Questions

- [wip] **GPU autonomy** — DMA, vertex sequencer, VGA
- ULX3S port for higher resolution rendering
- BVH ray traversal ISA extensions (path tracing)
- GONZALES path tracer integration

**Open questions — let’s discuss!**

- Can a TBDR pipeline scale to multi-core tiles?
- Minimal ISA for hardware BVH traversal?
- How to best expose Borg via Mesa Vulkan?

## Funding



NGI ZERO

Borg is funded by the **NGI0 Commons Fund**, established by NLnet Foundation with financial support from the European Commission’s Next Generation Internet programme (Grant Agreement No. 101135429, DG CONNECT), and by the Swiss State Secretariat for Education, Research and Innovation (SERI).

## Selected References

- M. Bell, *TinyQV: A RISC-V Core for Tiny Tapeout*, GitHub, 2024.
- J. Bush, *Nyuzi Parallel Processor*, GPGPU-7, 2014.
- A. Taneja et al., *Vortex: Open Source RISC-V GPGPU*, 2021.
- Woop et al., *RPU: A Programmable Ray Unit*, SIGGRAPH, 2005.
- Bachrach et al., *Chisel: Constructing HW in a Scala Embedded DSL*, 2012.
- C. Wolf, *Yosys Open SYNthesis Suite*, 2012.
- M. Venn, *Tiny Tapeout: Open Access Silicon*, tinytapeout.com, 2023.
- SkyWater/IHP, *Open Source 130 nm PDKs*, 2020–2024.
- Berkeley HW Group, *Hardfloat: Chisel FPU Cores*, 2018.
- A. Wendleder, *LibreLane init in nixpkgs*, PR #471712, 2026.

## Learn More



Source Code



Book / Docs

[andreas.wendleder@proton.me](mailto:andreas.wendleder@proton.me)  
[orcid.org/0009-0008-8304-2818](https://orcid.org/0009-0008-8304-2818)

Development supported by AI-augmented pair-programming.

## Comparison with Related Open GPUs

|                | Borg   | Nyuzi   | Vortex       |
|----------------|--------|---------|--------------|
| Architecture   | TBDR   | GPGPU   | RISC-V GPGPU |
| ISA            | Custom | Custom  | RV32         |
| Tapeout        | ✓      | —       | —            |
| Open toolchain | ✓      | partial | partial      |
| Target         | iCE40  | Arria 5 | Xilinx       |